

What Makes a Rock Song a Hit? Audio Features and Lyrical Analysis Across Rock Subgenres

DSA 210 - Spring 2026

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This project will investigate what audio and lyrical characteristics define and differentiate rock subgenres on Spotify, and whether these traits predict a song's popularity. The main question is: Do lyrical themes and sentiment add predictive power beyond audio features alone? And whether lyrical traits outperform audio features as predictors of popularity.

Primary dataset: A Kaggle dataset of ~114,000 Spotify tracks spread across 114 genres (~1,000 songs each), of which ~8,000 are rock-labeled. Each track includes Spotify audio features (energy, valence, tempo, danceability,...) and a popularity score ranging from 0 to 100, providing a great variety between non-hit and hit tracks.

Enrichment: I'm planning to enrich the dataset with lyrical annotations generated by an LLM or a secondary dataset. I will be trying to fetch lyrics from the Genius API (realistically expecting 3,000-5,000 matches). Lyrics passed to the LLM will be annotated into sentiment, lyrical themes, mood and subgenre.

Data characteristics: ~8,000 rock songs across 8 subgenres. Audio features are continuous numeric variables. Popularity scores range 0-96 with a mean of ~35, offering a genuinely balanced spread.